

Using the Bayer Patterns to Detect Fake Images

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Abstract—Given the rapid advancements in image synthesis and deep generative models, the proliferation of high-quality fake image generation, manipulation, and dissemination has become pervasive. This trend poses a threat to the public and potentially harms individuals’ reputations and safety through the circulation of counterfeit images. Previous efforts in detecting fake images primarily relied on deep learning techniques, which are computationally expensive and require extended training periods. In this study, we introduce a novel method for fake image detection, which is termed “redemosaicing.” This method entails converting raw RGB images into color filter array (CFA) images based on the Bayer patterns. Subsequently, we demosaic these CFA images to reconstruct new RGB images, which are then compared with the originals. We experimented with our proposed method on several datasets of real and fake images to examine the feasibility of the proposed approach.

I. INTRODUCTION

The rapid progress in image processing and machine learning algorithms has significantly facilitated the creation, manipulation, and production of high-quality synthesized images. Notably, advancements in technologies such as generative adversarial networks (GANs) [1]–[3], autoregressive convolutional neural networks (CNNs) [4], [5], and diffusion networks [6]–[8] have led to the generation of images so realistic that they are indistinguishable from genuine photographs captured by cameras to the human eye. However, the same technologies that benefit us also pose a threat when exploited maliciously, particularly in cybercrime. Attackers can leverage these tools to create deceptively counterfeit images, leading to various forms of harm. For instance, Deepfake [9] is a new technology that enables the replacement of a source identity’s face with that of a target identity, maintaining the original facial expressions with high realism. This powerful tool can be easily exploited for nefarious purposes, such as counterfeiting images targeting specific individuals or events and circulating those synthetic images containing inappropriate content on social media platforms, resulting in significant social and ethical dilemmas or detrimental to a person’s reputation and safety. Therefore, fake image detection has emerged as an urgent

imperative, necessitating the development of effective and efficient image forgery detection systems.

A conventional approach to detecting fake images is to train a new deep neural network as a binary classifier. The prior study [10] assessed techniques for identifying face manipulation, including CNN-based models for replacing faces. They demonstrated that basic classifiers could detect fakes from the same model but did not explore the generalization problem across datasets [11]. Several studies [12]–[14] have noted that classifiers trained on one GAN architecture struggle when applied to others due to variations in datasets, network designs, loss functions, and image preprocessing methods. Additionally, [15] found that simple classifiers could identify images from an image translation network [16] but did not investigate cross-model applicability.

A fundamental challenge that arises in directly training this kind of classifier is the difficulty in identifying common features distinguishing fake images generated by different generative models. These images, originating from diverse generative models, undergo varied transformations or processes of synthesis, which lack consistent discriminative characteristics. To address such issues, existing studies [17], [18] employed deep learning-based approaches incorporating contrastive loss for fake image detection. These methods leverage state-of-the-art GANs to produce pairs of fake and real images, forming the input dataset. Then, the goal is to find the typical features present in synthesized images from different generative models by training a classifier to identify generated images. Another strategy proposed by [19] involves combining co-occurrence matrices and deep learning techniques for fake image detection. This approach extracts co-occurrence matrices from the pixel domain’s three color channels and trains a CNN model accordingly. Moreover, an analysis conducted by [20] in the frequency domain revealed artifacts in generated images, facilitating the identification of fake images.

Prior research in detecting fake images predominantly relies on deep learning networks, which are time-consuming and computationally expensive in nature. In this course project, we



Fig. 1. The four Bayer patterns

attempt to leverage the fact that most cameras use the Bayer pattern [21] as the color filter array to differentiate between camera-captured images and computer-generated images.

In this project, a real image is defined as a photograph captured by a camera in the physical world, where the camera’s photosensor detects the intensity of the light and captures information necessary for image formation. This process involves converting varying light wave attenuations (encountered as light passes through or reflects off objects) into signals, represented as small bursts of current conveying pertinent information. Color image sensors come in several main types, each distinguished by its color-separation mechanism. Our approach focuses on integral color sensors [22], which incorporate a color filter array (CFA) atop a single monochrome CCD or CMOS image sensor. Among the most prevalent CFA patterns is the Bayer pattern, characterized by a checkerboard layout comprising two green, one red, and one blue for each four-pixel set. The four Bayer patterns are shown in Fig. 1.

As each pixel in the Bayer pattern image captures only one color (e.g., green), the absent color values (e.g., red and blue) for each pixel are estimated through interpolation using neighboring pixels [23]. This interpolation process, also known as “demaicing,” aims to reconstruct missing color for all RGB channels.

The rest of the report is organized as follows: Section II describes the proposed method to differentiate fake images from real images, the image comparison metrics, the image datasets and the demosaicing algorithms used in the experiments. Section III shows the results from the experiments. Section IV summarizes the findings of the project and suggests future research directions.

II. METHODOLOGY

In this section, we provide an overview of the proposed method to detect fake images, the demosaicing algorithm used in our proposed method, the datasets and the image comparison metrics used in the experiments.

A. Proposed Method

This subsection delineates our “redemosaicing” methodology, as illustrated in the accompanying Fig. 2. Initially, an RGB image is given as input. For each of the four Bayer patterns, we apply the “redemosaic” process to the original image:

- Perform the “redemosaicing” algorithm to transform the RGB image into a CFA image based on the selected Bayer pattern. For each pixel in the RGB image, we only keep the color channel specified by the given Bayer

pattern and remove the other two color channels. The resulting CFA image is a single-channel grayscale image.

- The output CFA image is demosaiced to reconstruct a new RGB image.
- Finally, we compare the newly obtained image and the original image using image comparison metrics.

For each image comparison metric, four image comparison results are calculated by comparing the original image and the four “redemosaiced” images. We record the comparison results and calculate the standard deviation of the four values. The Bayer pattern that gives the least difference between the original and the “redemosaiced” image is predicted to be the Bayer pattern used by the camera to take the image. We hypothesize that because fake images are not taken by cameras or altered after being taken by a camera, they do not have an underlying CFA image with a specific Bayer pattern. The image comparison results between the original image and all four “redemosaiced” images will be very similar. For real images taken by cameras, when they are “redemosaiced” by the correct Bayer pattern, the difference between the original image and the “redemosaiced” image will be small; when they are “redemosaiced” by the wrong Bayer patterns, the difference between the original image and the “redemosaiced” images will be larger. Therefore, the standard deviation of the real images’ comparison results will be relatively larger than that of the fake images. Thus, we experiment with the proposed “redemosaic” method on the datasets of real and fake images and collect the distributions of standard deviations of these image comparison results.

After getting the distributions of standard deviations of the image comparison metrics of the real and fake images, we try to take certain percentiles of the distributions of the real image and use them as hard thresholds. For an image for which we want to test if it is real or fake, we apply the same “redemosaic” process to the image, find the standard deviation of the four image comparison results, and check if the value falls into the range specified by the thresholds. We predict the image is real if the standard deviation value is within the range set by the thresholds.

B. Demosaicing Algorithm

The primary demosaicing algorithm used in our experiments is the method developed by Malvar et al. [24]. The Malvar-He-Cutler demosaicing method is a gradient-corrected form of bilinear interpolation. It improves bilinear taking by exploiting the correlation between different color channels. The design of this method is based on the notion that luminance components are much stronger than chrominance components at edges. In essence, a sharp change in a color channel correlates to a sharp change in luminance in the same pixel; thus, when interpolating values for missing color channels, the value for the color captured at a given pixel location holds pertinent information that improves the accuracy of the interpolated colors. Therefore, during the process of interpolation, a portion of the luminance change in one color channel must be factored into the bilinear interpolation for improved accuracy [24].

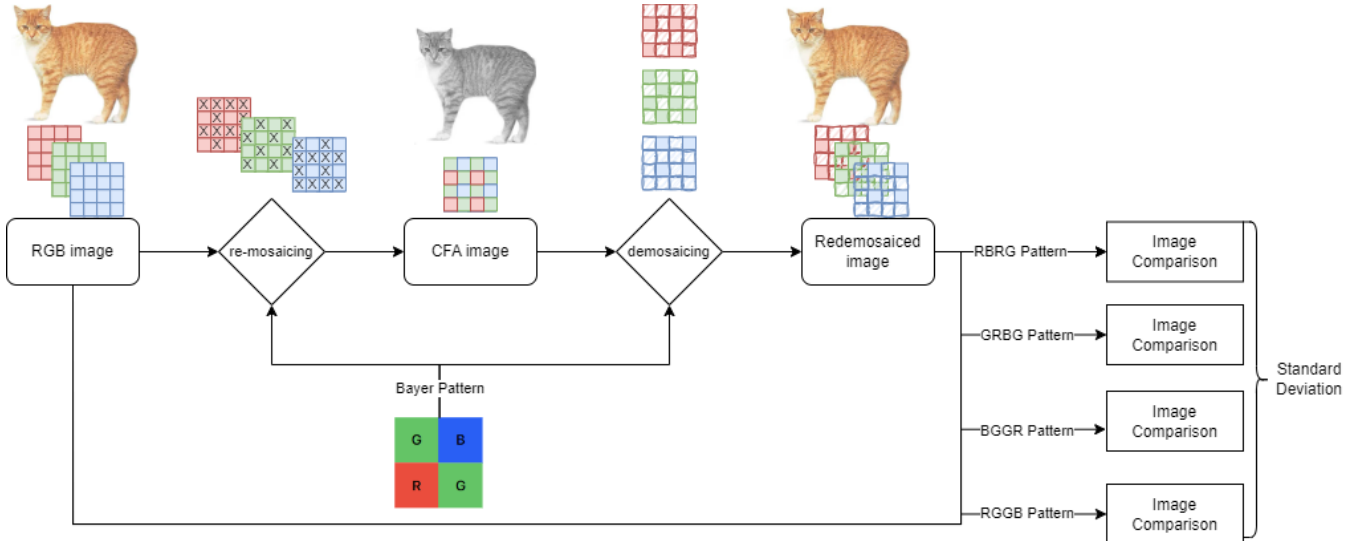


Fig. 2. Framework of “redemosaicing” methodology

C. Datasets

For our experiments, we used real images captured by a camera and artificially generated images to test our proposed method. These images were sourced from four datasets: RAISE [25], DiffusionDB [26], GenImage [27] and Landscape PhotoReal datasets.

1) *RAISE Dataset*: The Raw Images Dataset (RAISE) was created as a general-purpose dataset in the field of image forensics. The RAISE dataset consists of 8,156 RAW uncompressed images that have never been retouched or processed. The images in this dataset were captured using three cameras: Nikon D40, Nikon D90 and Nikon D7000. All the images have high resolutions (3008×2000 , 4288×2848 , and 4928×3264) and are stored in either Compress Raw (12-bit) or Lossless Compress (14-bit) format. For our experiment, we used 1000 images from the RAISE 1000 dataset to test our algorithms. The RAISE dataset provided an excellent resource for our experiments, as it contained high-resolution, camera-native images that have unaltered CFA patterns. Thus, we could use these images as “real images” in our experiments.

2) *DiffusionDB Dataset*: The DiffusionDB dataset is a massive collection of high-resolution images generated from written prompts using Stable Diffusion [28]. It is a large-scale dataset made up of 14 million user-generated images. The images in this dataset were created using 1,819,808 unique prompts with the hyperparameters selected by users. For our experiment, we used 1000 images from this dataset as a fake image dataset to test the validity of our proposed method.

3) *GenImage Database*: This dataset was created to facilitate the accurate and effective detection of AI-generated images. The GenImage dataset contains a total of 2,681,167 images, divided between real images and artificially generated images. The real images are taken from a subset of the ImageNet database. To generate the fake images, the creators used eight cutting-edge generative models as stated in the



Fig. 3. Sample images from the Landscape PhotoReal dataset.

reference paper [27], namely BigGAN [29], Stable Diffusion (V1.4 & V1.5) [28], Midjourney [30], ADM [31], Wukong [32], VQDM [33] and GLIDE [34]. The generated fake images are of different resolutions based on the model used in creating them. For this project, we only used 44,000 fake images in the GenImage dataset.

4) *Landscape PhotoReal Dataset*: In addition to the DiffusionDB dataset and the GenImage dataset, we created another dataset of fake images ourselves. We used the “Landscape_PhotoReal” [35] model, which is a finetuned Stable Diffusion 1.5 [36], to generate 1,000 images of size $1,024 \times 768$. The goal is to generate images of scenes of photo-realistic environments. We used prompts like “leaf,” “mountain,” “beach,” “snow,” and “cabin” to generate scenes that commonly appear in real photographs. We manually filtered

out generated pictures that contained erotic content and looked extremely fake. Fig. 3 shows some sample images in this generated dataset.

D. Image Comparison Metrics

For this project, we have chosen to utilize three full-reference objective image quality assessment metrics to compare the original images with the “redemosaiced” images, which are PSNR, SSIM, and Delta E.

The peak signal-to-noise ratio (PSNR) is the ratio of the maximum (i.e., peak) possible signal power to the power of the distorting noise that affects the quality of its representation [37]. In image processing, the comparison is done between a reference/original image and a test image. Both images must have the same dimension and are compared on a pixel-by-pixel basis [38].

The structural similarity index measure (SSIM) is an image quality metric developed as a perception-based model [39]. This metric is based on the idea of structural information, which posits that pixels that are located close to each other possess strong interdependencies, and these dependencies contain pertinent information about the structure of the object in the image. SSIM breaks down image distortion into three relatively independent components: luminance, contrast, and structure [38], [39]. The SSIM index value is then calculated using these three components.

Delta E (often denoted as ΔE , E^* or dE) is a full reference quality metric created by the International Commission on Illumination (CIE) that quantifies the distance or difference between two colors in the CIELAB color space. It is a quality metric that focuses on color accuracy between the reference image and the test image. In general, the lower the delta E values, the better the color accuracy [40]. Delta E is measured on a scale from 0 to 100, where 0 means no color difference, and 100 represents absolute color distortion. There are multiple versions of the Delta E algorithm created by CIE: dE76, dE94 and dE00 (also known as CIE2000). For this work, we used the CIE2000 version of the delta E [41], [42].

III. EXPERIMENT RESULTS

With the proposed method, for each image, we “redemosaic” the image with four Bayer patterns and calculate the image comparison result for each of the four pairs of original and “redemosaiced” images. We record the standard deviation of the four values for each image in a dataset, and for the entire dataset, we collect the distributions of the standard deviation values. Then, we set specific quantiles of the distributions of the real image dataset as thresholds. To detect if a given image is fake, we apply the same “redemosaic” process to it and calculate four image comparison metrics based on the original image and the “redemosaiced” images. We take the standard deviation of the four values and check if it falls into the range determined earlier. If so, we predict that it is a real image. Otherwise, we predict it to be a fake image.

Figure 4 shows the distributions of standard deviations of the image comparison metrics of the RAISE 1000 dataset

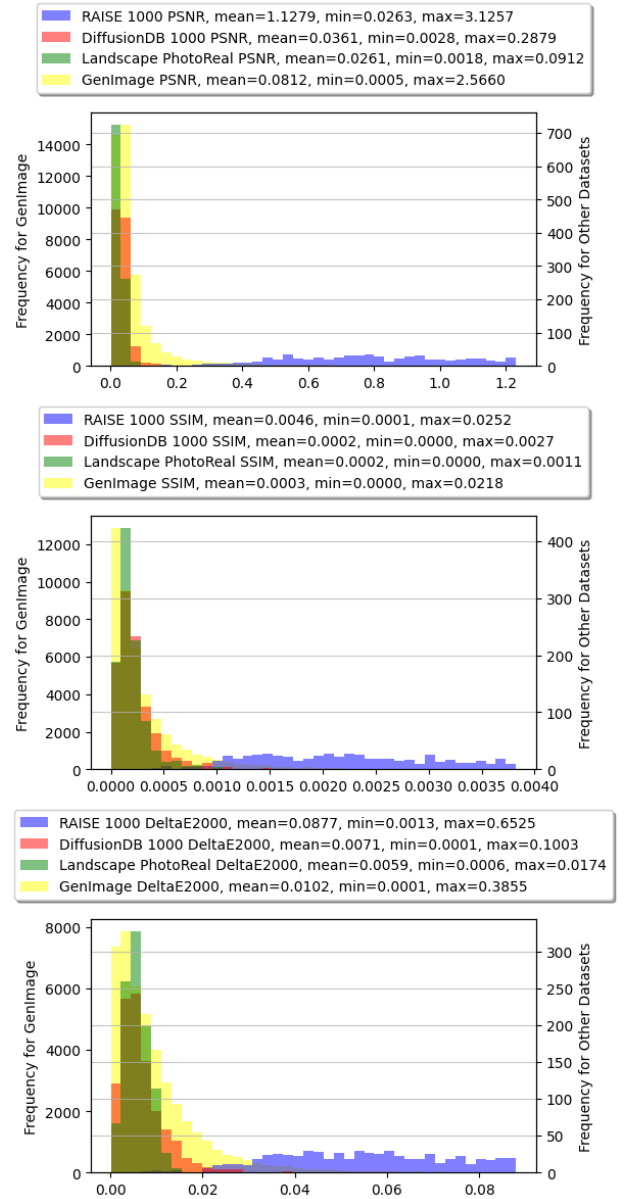


Fig. 4. The distributions of the standard deviations of image comparison metrics of the RAISE 1000 dataset, the GenImage dataset, the Landscape PhotoReal dataset, and the DiffusionDB 1000 dataset.

[25], the Landscape PhotoReal dataset, the GenImage dataset, and the DiffusionDB 1000 dataset [26]. In these histograms, the horizontal axis represents the standard deviation of the comparison metrics, and the vertical axis shows how many images whose comparison scores’ standard deviations fall into a range of the comparison scores represented by the bin.

As shown by the plots, the standard deviations of the PSNR, SSIM and Delta-E values of the real dataset are, in general, much higher than those of the fake images. We hypothesize that because the fake images were not taken by cameras using the Bayer pattern in the first place, the comparisons between the original image and the “redemosaiced” images will be similar no matter what Bayer pattern is used in the “rede-

TABLE I
PREDICTION ACCURACIES OF USING THE PERCENTILES OF
DISTRIBUTIONS OF STANDARD DEVIATIONS OF COMPARISON METRICS AS
THRESHOLDS.

	PSNR	SSIM	Delta E
<i>DiffusionDB Prediction Accuracies</i>			
(0, 100)-Percentile Range	0.35	0.16	0.06
(1, 99)-Percentile Range	1.00	0.97	0.92
(2, 98)-Percentile Range	1.00	0.99	0.98
(3, 97)-Percentile Range	1.00	0.99	0.98
(5, 95)-Percentile Range	1.00	0.99	0.99
(10, 90)-Percentile Range	1.00	1.00	0.99
<i>GenImage Prediction Accuracies</i>			
(0, 100)-Percentile Range	0.25	0.27	0.06
(1, 99)-Percentile Range	0.96	0.89	0.79
(2, 98)-Percentile Range	0.96	0.93	0.90
(3, 97)-Percentile Range	0.97	0.94	0.92
(5, 95)-Percentile Range	0.97	0.95	0.94
(10, 90)-Percentile Range	0.98	0.97	0.96
<i>Landscape PhotoReal Prediction Accuracies</i>			
(0, 100)-Percentile Range	0.57	0.15	0.01
(1, 99)-Percentile Range	1.00	0.98	1.00
(2, 98)-Percentile Range	1.00	0.99	1.00
(3, 97)-Percentile Range	1.00	1.00	1.00
(5, 95)-Percentile Range	1.00	1.00	1.00
(10, 90)-Percentile Range	1.00	1.00	1.00

mosaic” process. However, for the real images, the difference between the original image and the “redemosaiced” image will be greater if the Bayer pattern used in the “redemosaic” process is not the pattern used by the camera.

After inspecting the image comparison results manually, we found that for the real images, the image comparison results of the “BGGR-redemosaiced” and “RGGB-redemosaiced” images are usually similar; the image comparison results of the “GRBG-redemosaiced” and “GBRG-redemosaiced” images are usually similar. However, the image comparison results for these two “Bayer pattern groups” are usually very different, making the standard deviation across all four image comparison results larger. For fake images, the four image comparison results are close to each other as expected. Therefore, even though the Bayer pattern giving the lowest image comparison difference may not actually be the correct Bayer pattern used by the camera, the standard deviation across all four Bayer patterns’ comparison results can still be used to detect if an image is a fake image or not.

Table I shows the prediction accuracies of using different percentiles of the distributions as thresholds to identify fake images from the Landscape PhotoReal dataset, the DiffusionDB dataset and the GenImage dataset [27]. If we use the maximum and the minimum values of comparison results of the real images as the thresholds, which corresponds to the (0, 100)-percentile range, the performance is bad as there are some outliers in the RAISE 1000 dataset whose standard deviations of the comparison scores are very small. Tuning the percentile range smaller will increase the accuracy of detecting fake images at the cost of more real images being wrongly classified as fake images as well. In general, we found that using the standard deviations of the PSNR comparisons to detect fake images is most effective. For the DiffusionDB

dataset and the Landscape PhotoReal dataset, this method can achieve near near-perfect detection rate. The GenImage dataset is more challenging in that it contains images from different generative models of different styles. Using PSNR’s standard deviation values can still achieve about 97% accuracy.

IV. CONCLUSION AND FUTURE DIRECTIONS

In this project, we experimented with a novel method to detect fake images synthesized by generative models. The proposed approach takes advantage of the fact that most modern cameras use the Bayer pattern as the color filter array when taking images, and the images taken by such cameras need to be demosaiced for missing colors. The fake images synthesized by generative models do not go through such a process. Thus, we hypothesized that when we convert an RGB image taken by a camera back to a CFA image and re-apply the demosaic process, the difference between the original image and the “redemosaiced” image would be smaller if the correct Bayer pattern is used in the “redemosaic” process. The comparison results would be similar for the fake images without the underlying Bayer patterned CFA image, no matter what Bayer pattern is used in the “redemosaic” process. Hence, we can detect if an image is fake by calculating the standard deviation of the image comparisons between the original and the four “redemosaiced” images, expecting the standard deviation of the four image comparisons to be greater for real images.

The experiment results showed very promising results on all three fake image datasets on which we experimented. Despite some real images not having large standard deviations of image comparison results, most real images’ image comparison standard deviation is significantly greater than that of fake images, making it easy to separate the distributions of standard deviations of image comparisons of fake and real images. In practice, we can either empirically analyze the distributions of the real images and manually select specific percentiles of the distributions as classification thresholds or use machine learning classifiers to learn such decision boundaries.

One major limitation of the proposed approach is that it only works for real images taken by cameras with Bayer pattern color filter arrays, and it will not work for real images whose underlying Bayer pattern is broken by image alterations like compression and resizing. As most images found online are compressed or resized for file size and formatting considerations, our proposed method may not apply to those images.

Finally, as shown by the experiment results, many of the real images also have small standard deviations of image comparison results. The proposed method will fail to correctly classify these images. Future works can delve further into what contributes to the difference in the standard deviations and, in general, what kind of images will have small standard deviations of image comparisons. Such a study may provide us with more insight into the limitations of the proposed method.

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